

1. Build a Docker Container from a Custom Dockerfile

Project Name: program_devops_1

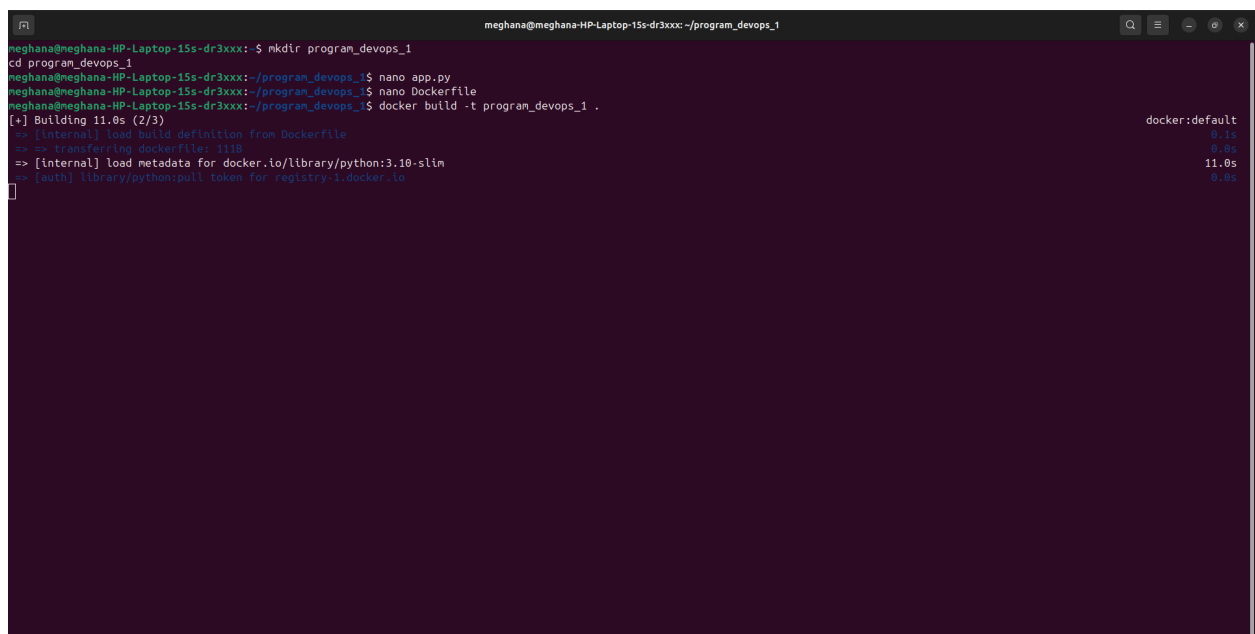
program_devops_1/

```
|
|— app.py
|— requirements.txt
|— Dockerfile
```

Step 1 – Create Project Directory

```
mkdir program_devops_1
```

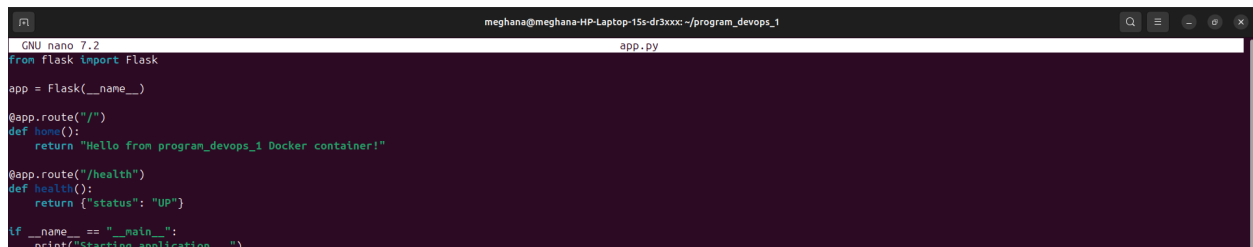
```
cd program_devops_1
```

A terminal window with a dark purple background and white text. The window title is 'meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_1'. The terminal shows the following commands and output:

```
meghana@meghana-HP-Laptop-15s-dr3xxx: $ mkdir program_devops_1
cd program_devops_1
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_1$ nano app.py
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_1$ nano Dockerfile
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_1$ docker build -t program_devops_1 .
[+] Building 11.0s (2/3)
=> [internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 11kB 0.0s
=> [internal] load metadata for docker.io/library/python:3.10-slim 11.0s
=> [auth] library/python:pull token for registry-1.docker.io 0.0s
docker:default
```

Step 2 – Create Application File (app.py)

Step 3 – Create requirements.txt



```
GNU nano 7.2 app.py
from flask import Flask

app = Flask(__name__)

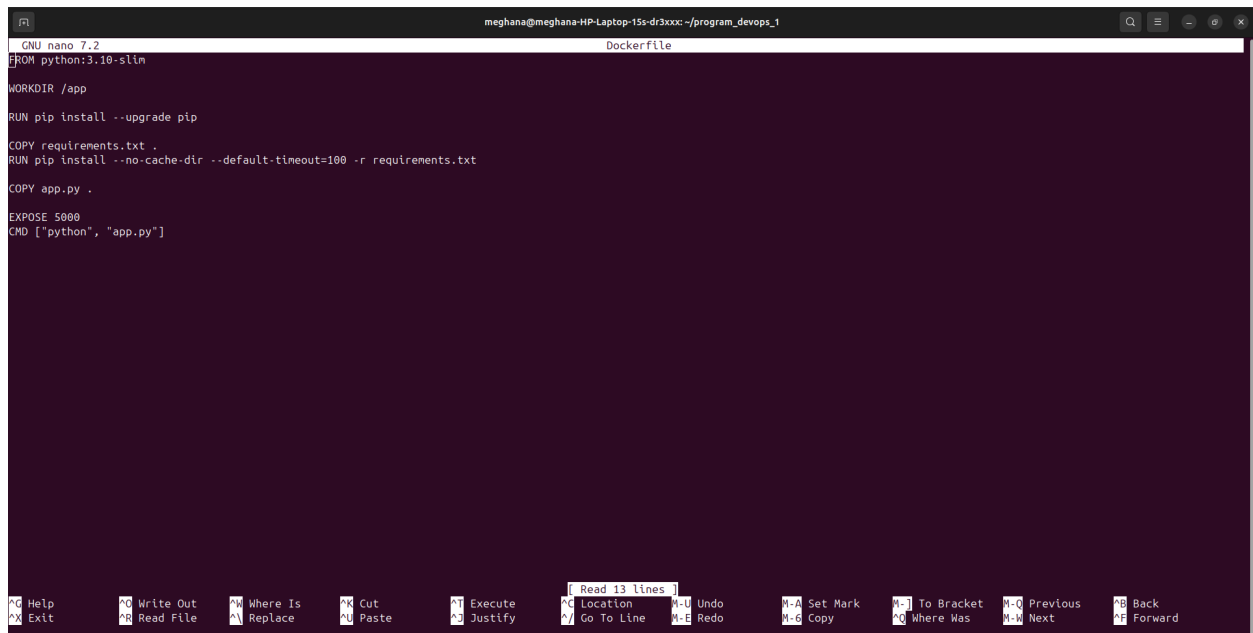
@app.route("/")
def home():
    return "Hello from program_devops_1 Docker container!"

@app.route("/health")
def health():
    return {"status": "UP"}

if __name__ == "__main__":
    print("Starting application...")
```

Step 4 – Create Custom Dockerfile

nano DockerFile



```
GNU nano 7.2 Dockerfile
FROM python:3.10-slim

WORKDIR /app

RUN pip install --upgrade pip

COPY requirements.txt .
RUN pip install --no-cache-dir --default-timeout=100 -r requirements.txt

COPY app.py .

EXPOSE 5000

CMD ["python", "app.py"]
```

Step 5 – Build Docker Image

docker build -t program_devops_1 .

```

meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_1$ nano requirements.txt
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_1$ nano Dockerfile
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_1$ docker build -t program_devops_1 .
[+] Building 158.2s (12/12) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 257B
=> [internal] load metadata for docker.io/library/python:3.10-slim
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/6] FROM docker.io/library/python:3.10-slim@sha256:7b68a5fa7cf8d20b4cedb1dc9a134fdd394fe27edbc4c2519756c91d21df2313
=> [internal] load build context
=> => transferring context: 79B
=> CACHED [2/6] WORKDIR /app
=> [3/6] RUN pip install --upgrade pip
=> [4/6] COPY requirements.txt .
=> [5/6] RUN pip install --no-cache-dir --default-timeout=100 -r requirements.txt
=> [6/6] COPY app.py .
=> exporting to image
=> => exporting layers
=> => writing image sha256:5bbd194bf17ef9e511b92fc375414c80a2a011859446eda3d56427efad97c32
=> => naming to docker.io/library/program_devops_1
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_1$

```

Step 6 – Verify Docker Image

docker images

Step 7 – Run Docker Container

docker run -p 5000:5000 program_devops_1

Step 8 – Application Output

Starting application...

Running on http://127.0.0.1:5000



